25900250

MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP

(HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCBTG202 –

FUNDAMENTALS OF MOLECULARBIOLOGY

(2024 ADMISSION ONWARDS)

SCHEME

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
One word or sentence

Answer any 10 questions Each question carries 2 marks

1. Mention two proteins that are associated with bacterial nucleoid organization [U]/[CO1]
   Any two proteins- 2 marks
2. What is the function of DNA Polymerase? [K]/[CO2]
   Any two functions- 2 marks
3. What is the holoenzyme of RNA polymerase made up of ? [A]/[CO3]
   Core enzyme+ sigma factor - 1.5 mark, components of core enzyme- .5 marks
4. Name the two types of chromatin. [K]/[CO1]
   Heterochromatin, Euchromatin- 2 marks
5. Which enzyme catalyzes the addition of poly(A) tail to mRNA? [K]/[CO3]
   Poly A polymerase- 2 marks
6. Recall the DNA form that appears in dehydrated conditions. [K]/[CO1]
   A DNA- 2 marks
7. Identify the role of the promotor in gene regulation. [U]/[CO3]
   Serves as the binding site for RNA polymerase and regulates the initiation of transcription- 2 marks.
8. Mention two ways in which translation can be regulated in eukaryotes. [K]/[CO3]
   Any two ways-(By controlling initiation factors, by modulating ribosome binding to mRNA, Antibiotics etc.) - 2 marks
9. Name the scientists who discovered the structure of DNA, and the year. [K]/[CO1]
   James Watson and Francis Crick- 1 mark, 1953- 1 mark

10. In negative regulation, what happens when the repressor is bound to the operator? [U]/[CO3]
    Binding of the repressor to the operator **prevents transcription** by blocking RNA polymerase- 2 marks.
11. Name the strains of Streptococcus pneumoniae used in Griffth's experiment. [K]/[CO1]
    R (Rough) strain, S (Smooth) strain- 2 marks
12. Name the host plant commonly infected by TMV. [K]/[CO1]
    Tobacco plant (members of the family Solanaceae) - 2 marks

[2x10 = 20]

Part B

Short Essay-Answer any 5 questions

Each question carries 4 marks

13. Differentiate between mRNA and tRNA in terms of structure and function [U]/[CO1]
    Structure with diagram – 2 marks, Function – 2 marks (Minimum 4 Points)
14. Explain photoreactivation as a direct DNA repair mechanism [U]/[CO2]
    Mechanism – 3 marks, Diagram - 1 mark
15. Describe the role of reverse transcriptase in HIV replication [U]/[CO3]
    RNA Virus/ retro virus – Introduction – 1 mark, Reverse transcription steps – 3 marks
16. Why is DNA replication described as semi conservative? [K]/[CO2]
    Semi- conservative replication- definition - 1 mark, Stahl and Meselson Experiment – 3 marks
17. Write a short note on glycosylation of proteins and its biological significance [U]/[CO3]
    Glycosylation-Process – 3 marks, Significance – 1 mark
18. Differentiate between structural gene and regulatory gene [U]/[CO1]
    Structural gene -2 marks (4 points), Regulatory gene – 2 mark (4 points)
19. Why is lac operon considered as inducible operon? Relate to cAMP and CAP [U]/[CO3]
    Inducible operon – 1 mark, Catabolite repression – 2 marks , Diagram – 1 mark

[4x5 = 20]

Part C

Essay type questions

Answer any 1 questions -Each questions carries 10 marks

20. With a neat labelled diagram explain the replication of DNA in eukaryotes. [U]/[CO2]
    Introduction – 1 mark, Enzymes involved in DNA replication – 2 marks, Steps in DNA Replication – 3 marks, Diagram – 4 marks
21. Describe in detail hoe translation occurs in prokaryotes ? [K]/[CO3]
    Intoduction – 1 mark, Steps in translation – 7 marks, Diagram – 2 marks

[10x1 = 10]